

3(a)	<table><tr><th>feature</th><th>artery</th><th>capillary</th><th>vein</th></tr><tr><td>smooth muscle</td><td>✓</td><td>X</td><td>✓</td></tr><tr><td>endothelium</td><td>✓</td><td>✓</td><td>✓</td></tr><tr><td>tunica media</td><td>✓</td><td>X</td><td>✓</td></tr></table> ;;; one mark per correct row if 0 or 1 mark, check columns to allow max 2	feature	artery	capillary	vein	smooth muscle	✓	X	✓	endothelium	✓	✓	✓	tunica media	✓	X	✓	3
feature	artery	capillary	vein															
smooth muscle	✓	X	✓															
endothelium	✓	✓	✓															
tunica media	✓	X	✓															
3(b)(i)	labelled arrows for oxygen, leaving alveolar space / entering capillary lumen and carbon dioxide, leaving capillary lumen / entering alveolar space ;	1																
3(b)(ii)	any four from: 1 blood arriving (in alveolar capillaries), is deoxygenated / has low(er) partial pressure of oxygen ; A low, concentrations / levels, of oxygen or oxygen-poor 2 oxygen taken up by haemoglobin (from plasma) / formation of oxyhaemoglobin ; 3 <i>idea that</i> newly oxygenated blood is, constantly / continually, removed ; A oxygenated blood passed to pulmonary vein A oxygen that diffuses, from alveoli / into blood, is transported away (by blood flow) 4 (because of) many / large network of capillaries ; <i>context is alveolar capillaries</i> 5 ventilation / breathing / inhalation and exhalation / AW, qualified ; e.g. allows (more) oxygen in (to alveolar space) <i>idea of</i> constantly replaces alveolar air 6 AVP ; e.g. ventilation maintains a large concentration difference between alveolar space and capillary	4																
3(c)	pulmonary vein ; semilunar ; A aortic <u>and</u> pulmonary coronary ;	3																